Meet 0

October 19, 2026

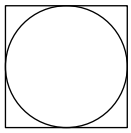
Geometry

You have 20 minutes to complete 6 problems. There is no penalty for guessing. Answers are only valid if they are written in the below table.

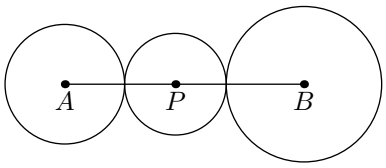
First and last name	

p	Answer	p	Answer
1		4	
2		5	
3		6	

1. Alice and Bob are running a race around a square track with side length 100 meters. Alice runs around the full length of the square, but Bob cuts corners by instead running around the square's inscribed circle. How much further does Alice run than Bob, rounded to the nearest integer?

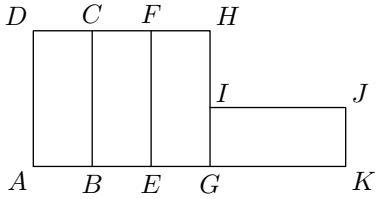


2. Circles X and Y with centers A and B have radii 20 and 26, respectively. Let Z be the circle externally tangent to X and Y with center P on $\overline{AB}$. Suppose $\overline{AP} = 37$. Compute the length of $\overline{AB}$.

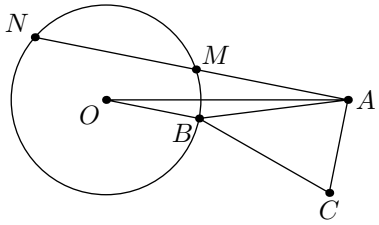


3. Parallelogram $ABCD$ satisfies $CD > BC$. Point E lies on side AB so that $AD = DE$, and point F lies on side CD so that $BF = FC$. If $DE \parallel BF$, what is $\angle BAD$, in degrees?

4. In the figure below, $ABCD$, $BEFC$, $EGHF$, and $IGKJ$ are congruent rectangles, so that I lies on segment GH. Points K, I, and D are collinear. The ratio $\frac{AD}{AB}$ is $\frac{a+\sqrt{b}}{c}$ for some positive integers a, b, and c, where b is not divisible by the square of any prime. What is $a+b+c$?



5. Let ω be a circle with radius 2 centered at O, and let A be a point outside ω. A line ℓ passing through A intersects ω at M and N, where $AM < AN$. Another line ℓ_1 is constructed parallel to ℓ through O. Suppose ℓ_1 intersects ω at a point B on a different side of AO as M. Let C be the reflection of A across ℓ_1. The area of $\triangle ABO$ is 1, and the area of $\triangle ABC$ is 3. Compute $AM + AN$.



6. Let N be the unique positive integer such that

$$\frac{\cot\frac{\pi}{3}\cot\frac{\pi}{7}\cot\frac{\pi}{N}}{\cot\frac{\pi}{3}+\cot\frac{\pi}{7}+\cot\frac{\pi}{N}} = 1.$$

Compute N.